Multiple-choice questions

|!|BCEMA0180000001|!| ★★

When a bone is placed in hydrochloric acid, it becomes softer because

A. the collagen in the bone is denatured.

B. the bone marrow is destroyed.

C. the calcium salts in the bone are dissolved by the acid.

D. water is drawn out of the bone by osmosis.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0180000002|!|

What is the major inorganic constituent of bone?

A. sodium carbonate

B. calcium phosphate

C. magnesium phosphate

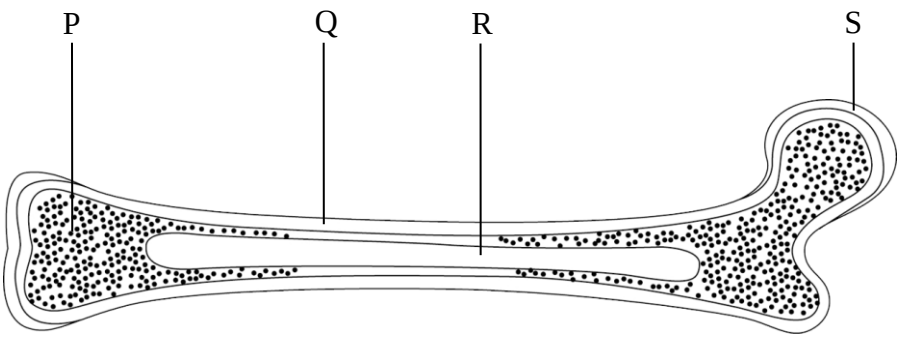
D. calcium chloride

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0180000003|!|

The diagram below shows the longitudinal section of a long bone.



© 2020 Aristo Educational Press Ltd.

Which part of the bone is the hardest?

A. P
B. Q
C. R
D. S

☐

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0180000004|!|

Which of the statements about bones is ***incorrect***?

A. They are the site of production of red blood cells.
B. They are for the storage of calcium.
C. They work with muscles to allow movement.
D. They can act as shock absorbers.

☐

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0180000005|!|

Which of the following is/are the function(s) of human skeleton?

(1) support and movement
(2) storage of fat
(3) breakdown of red blood cells

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

☐

##B##

Skill assessed	Mark
Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0180000006|!| ★

Bone is able to

(1) carry out respiration.
(2) store red blood cells.
(3) detect pain.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0180000007|!|

Which of the following is/are the function(s) of bone marrow?

(1) producing blood cells
(2) storing calcium
(3) storing fat

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMC0180000008|!|

Osteoporosis (loss of bone density) is caused by a deficiency of

A. vitamin A.
B. vitamin C.
C. potassium.
D. calcium.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0180000009|!| ★

Which of the following comparisons between bone and cartilage is/are correct?

(1) Bone is made up of living cells while cartilage is made up of dead cells.
(2) Bone contains blood vessels while cartilage contains no blood vessels.
(3) Bone contains more minerals than cartilage.

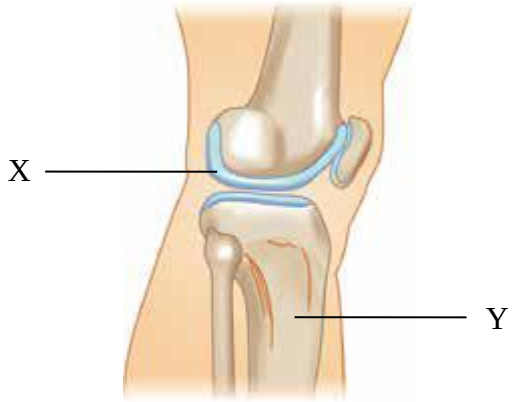
A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0180000010|!| ★

The diagram below shows part of the skeleton in the knee.



© 2020 Aristo Educational Press Ltd.

Which of the following comparisons between tissues X and Y is/are correct?

(1) X is softer than Y.
(2) Y contains more minerals than X.
(3) Y can repair itself while X cannot.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##B ##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0180000011|!| ★
Which of the following does ***not*** contain cartilage?
A. nose
B. trachea
C. oesophagus
D. vertebral column

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0180000012|!|
Which of the following is/are component(s) of the axial skeleton?

(1) cranium
(2) sternum
(3) pelvic girdle

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

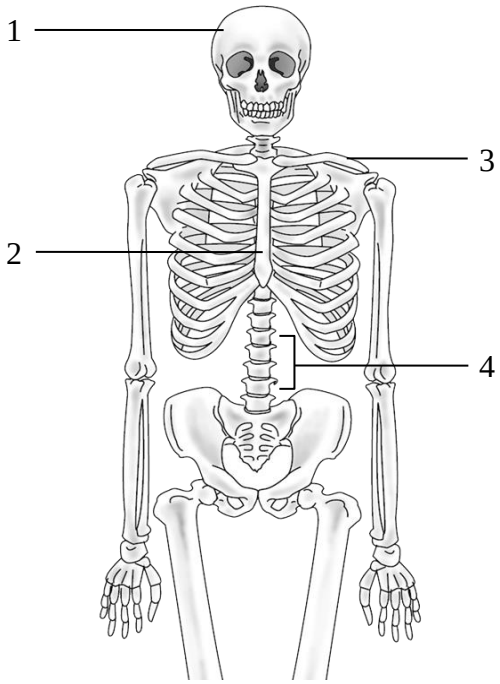
##B##

Skill assessed	Mark
Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0180000013|!| ★

The diagram below shows part of the human skeleton.



Which labelled parts are components of the axial skeleton?

A. 1 and 3 only

B. 2 and 4 only

C. 1, 2 and 4 only

D. 1, 2, 3 and 4

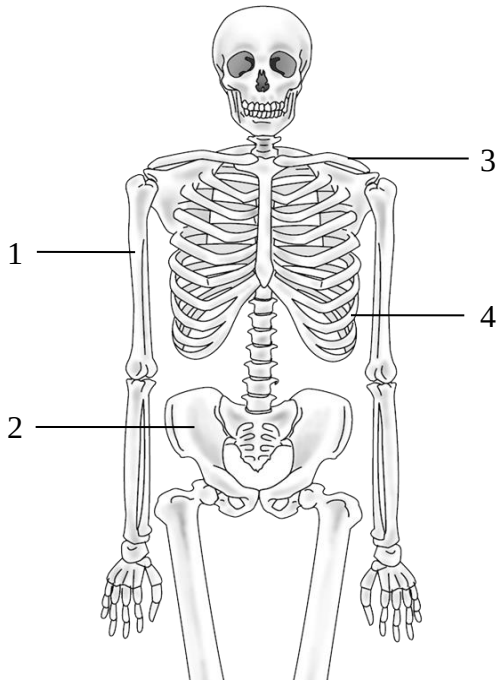
##C##

Skill assessed	Mark
Analysing data	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0180000014|!| ★

The diagram below shows part of the human skeleton.



Which labelled parts are components of the appendicular skeleton?

A. 1 and 2 only
B. 1 and 3 only
C. 1, 2 and 3 only
D. 1, 2, and 4 only

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0180000015|!|

What do the vertebral column and the rib cage protect?

	Vertebral column	***Rib cage***
A.	spinal cord	heart
B.	spinal cord	liver
C.	spinal nerves	heart
D.	spinal nerves	liver

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0180000016|!| ★

Which of the following is/are the function(s) of ribs?

(1) protecting the lungs
(2) supporting the body weight
(3) storing iron

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

☐

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0180000017|!| ★

Which of the following ***not*** a function of the rib cage?

A. giving shape to the body
B. protecting internal organs
C. assisting ventilation
D. storing red blood cells

☐

##D##

Skill assessed	Mark
Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0180000018|!| ★

Which of the following organs are protected by the pelvic girdle?

(1) uterus
(2) intestines
(3) urinary bladder

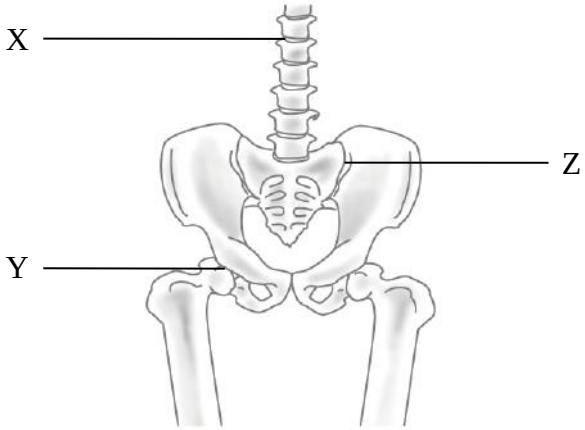
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0180000019|!| ★

The diagram below shows part of the human skeleton.



Which of the labelled joints allow movement between the bones?

A. X and Y only
B. X and Z only
C. Y and Z only
D. X, Y and Z

##A##

Skill assessed	Mark
Analysing data	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMC0180000020|!|

Which of the following is ***not*** made of living cells?

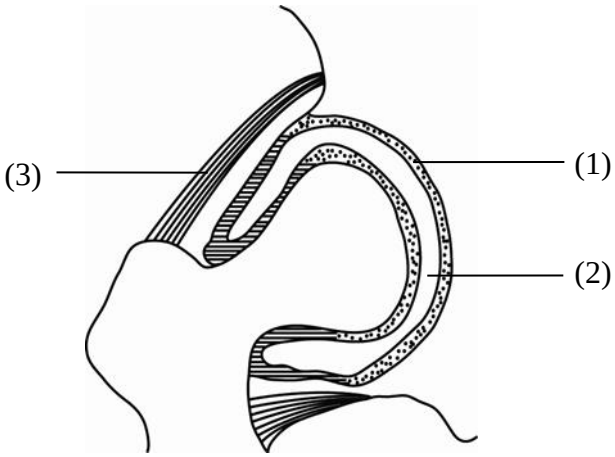
A. enamel
B. cartilage
C. bone
D. ligament

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0180000021|!|

The diagram below shows a movable joint in the human body.



Which of the labelled structures help reduce friction during movement?

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

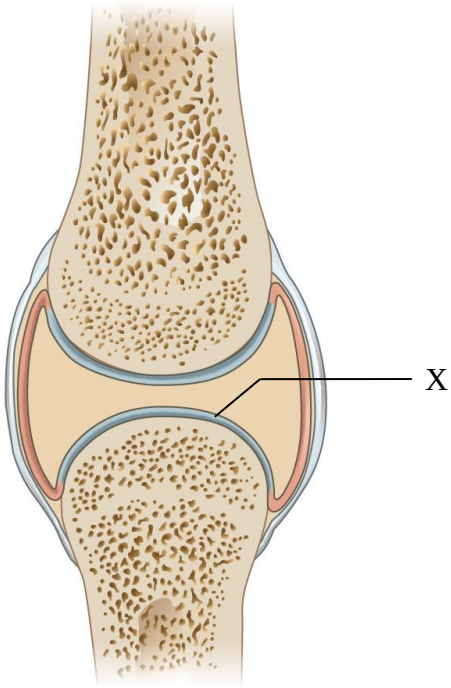
##A##

Skill assessed	Mark
Analysing data	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMC0180000022|!|

The diagram below shows a synovial joint.



Which of the following correctly describe(s) the functions of X?

(1) It acts as a shock absorber.
(2) It secretes synovial fluid.
(3) It produces blood cells.

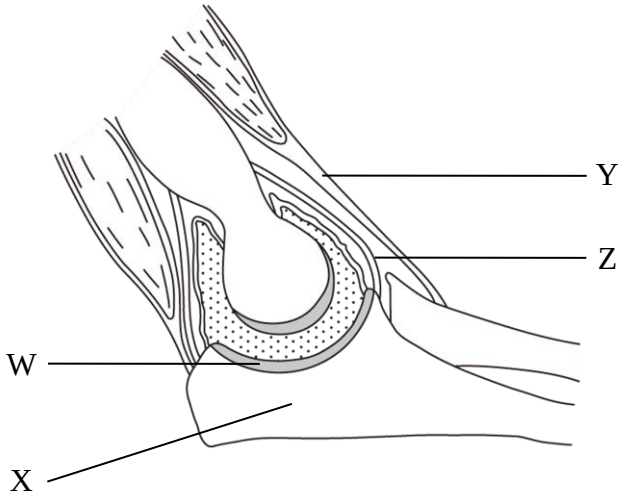
A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0180000024|!|

This diagram below shows an elbow joint and its associated structures.



Which of the labelled structures are elastic?

A. W and Y only
B. W and Z only
C. X and Y only
D. Y and Z only

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0180000025|!|

Which of the following statements about a tendon are correct?

(1) It connects a muscle to a bone.
(2) It transmits the pull of the muscle to the bone to cause movement.
(3) It is elastic.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0180000026|!| ★

Which of the following combinations correctly describes a tendon?

	Property	***Importance during movement***
A.	rich in glucose	to provide more energy for muscle contraction
B.	rich in collagen	to provide a smooth surface for reducing friction
C.	elastic	to allow the attached bones to move freely
D.	inelastic	to transmit pulling forces to the bone with minimum loss

☐

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0180000027|!|

Which of the following correctly describes the functions of ligaments and tendons?

	Ligaments	***Tendons***
A.	holding bones together	transmitting force to bones
B.	reducing friction at joints	protecting joints
C.	protecting joints	preventing dislocation of joints
D.	transmitting force to bones	holding bones together

☐

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0180000028|!|

What is the role of cartilage in the elbow joint?

A. preventing dislocation of the joint
B. connecting the triceps to the bones of the forearm
C. preventing wearing of bones during movement
D. secreting synovial fluid

☐

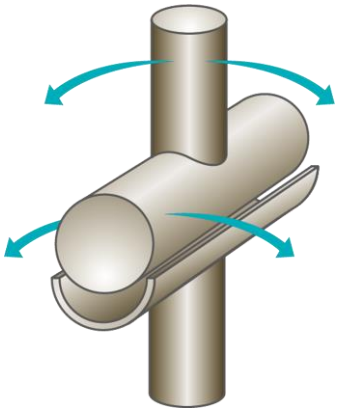
##C##

Skill assessed	Mark
Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMC0180000029|!|

The diagram below shows a model of a moveable joint.



Key: → direction of possible movements

Which of the following combinations correctly states the type of joint represented by the model and an example of this type of joint?

	Type of joint represented by the model	***Example***
A.	ball-and-socket joint	shoulder joint
B.	ball-and-socket joint	knee joint
C.	hinge joint	shoulder joint
D.	hinge joint	knee joint

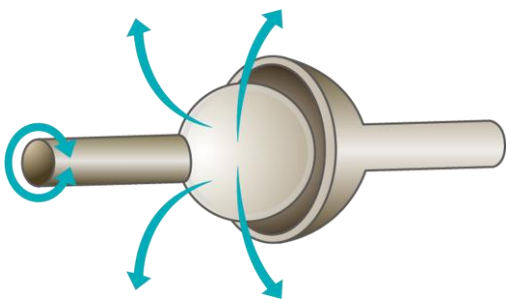
□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0180000030|!|

The diagram below shows a model of a moveable joint.



Key: → direction of possible movements

© 2020 Aristo Educational Press Ltd.

Which of the following combinations correctly states the type of joint represented by the model and an example of this type of joint?

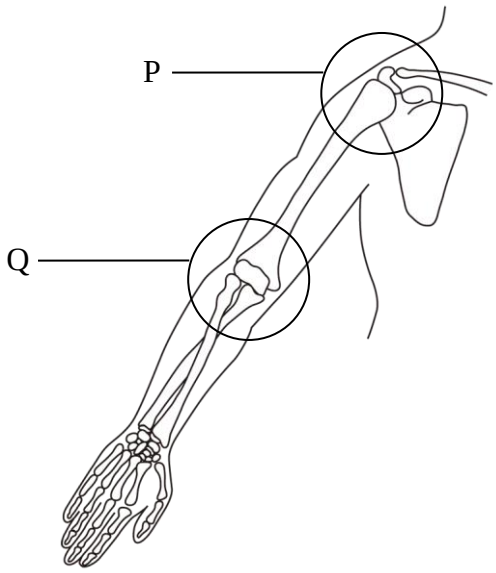
	Type of joint represented by the model	***Example***
A.	ball-and-socket joint	hip joint
B.	ball-and-socket joint	elbow joint
C.	hinge joint	hip joint
D.	hinge joint	elbow joint

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0180000031|!|

The diagram below shows the bones in a human arm.



What types of joints are P and Q?

	P	***Q***
A.	hinge joint	hinge joint
B.	hinge joint	ball-and-socket joint
C.	ball-and-socket joint	hinge joint
D.	ball-and-socket joint	ball-and-socket joint

##C##

Skill assessed	Mark
Analysing data	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMC0180000032|!|

Skeletal muscles are attached to bones by

A. cartilage.
B. tendons.
C. ligaments.
D. motor end plates.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0180000033|!|

In a joint, bones are held together by

A. cartilage.
B. tendons.
C. ligaments.
D. muscles.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0180000034|!|

Which of the following helps prevent dislocation?

A. synovial membrane
B. ligaments
C. tendons
D. cartilage

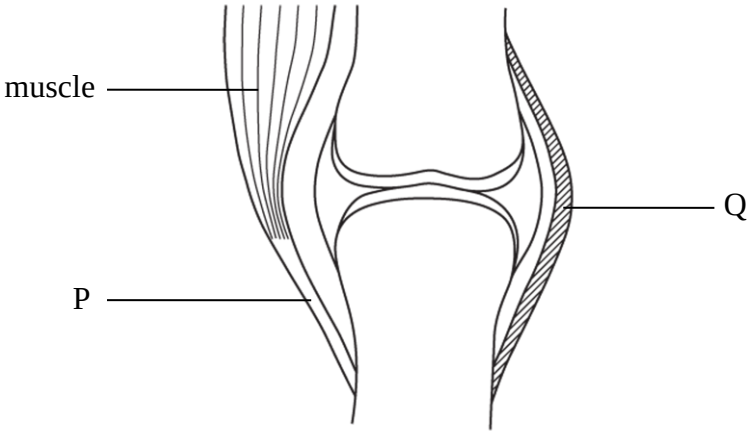
##B##

Skill assessed	Mark
Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0180000035|!| ★

The diagram below shows the structures associated with a joint.



Which comparison between structures P and Q is correct?

	P	***Q***
A.	living	non-living
B.	soft	rigid
C.	not elastic	elastic
D.	cannot stand a high tension	can stand a high tension

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0180000037|!| ★

Which of the following combinations correctly states the conditions of the arm muscles and the type of joint used in straightening the forearm?

	Triceps	***Biceps***	***Type of joint used***
A.	contracts	relaxes	ball-and-socket joint
B.	contracts	relaxes	hinge joint
C.	relaxes	contracts	ball-and-socket joint
D.	relaxes	contracts	hinge joint

□

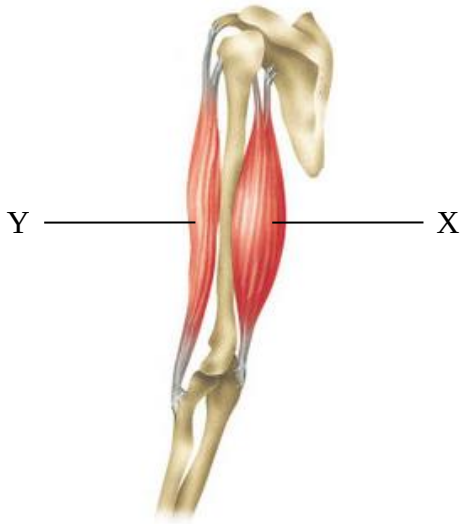
##B##

Skill assessed	Mark
Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0180000038|!| ★

The diagram below shows some bones and muscles in a human arm.



Which of the following statements about muscles X and Y are correct?

(1) Muscle X is the triceps.
(2) Muscles X and Y are antagonistic muscles.
(3) Muscle Y becomes shorter and thicker when the arm bends at the elbow.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0180000039|!|

Which of the following combinations correctly states the functions of cartilage, ligaments and synovial fluid in a synovial joint?

	Ligaments	***Cartilage***	***Synovial fluid***
A.	links bones to muscles	links bones together	lubricates the joint
B.	links bones to muscles	links bones together	maintains the shape of the joint
C.	links bones together	acts as a shock absorber	lubricates the joint
D.	links bones together	lubricates the joint	maintains the shape of the joint

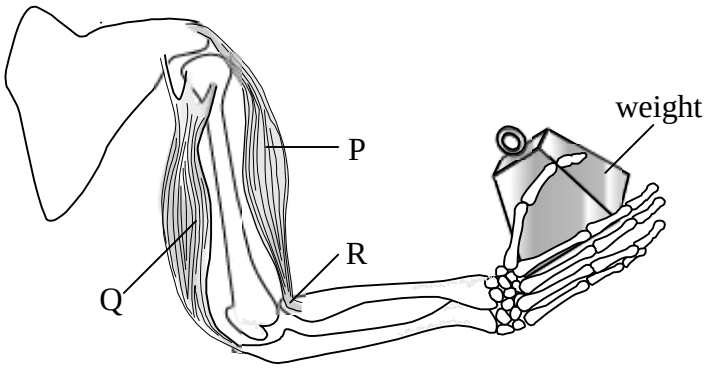
##C##

© 2020 Aristo Educational Press Ltd.

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0180000040|!| ★★

The diagram below shows a human arm lifting a weight.



When lifting the weight and maintaining the position as in the diagram,

A. P is contracting and R is lengthened.

B. P is contracting and R is under tension.

C. Q is contracting and R is shortened.

D. Q is contracting and R is slackened.

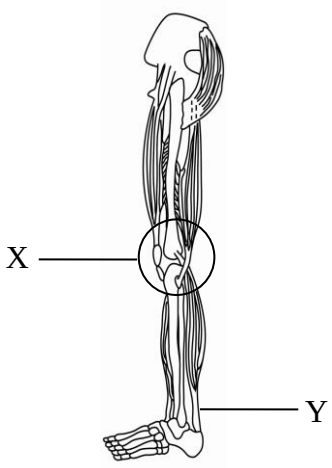
□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0180000041|!|

The diagram below shows the bones and muscles in a human leg.



© 2020 Aristo Educational Press Ltd.

Which of the following correctly identifies joint X and structure Y?

	X	***Y***
A.	hinge joint	ligament
B.	hinge joint	tendon
C.	ball-and-socket joint	ligament
D.	ball-and-socket joint	tendon

☐

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0180000042|!|

Which of the following is/are ball-and-socket joint(s)?

(1) elbow joint
(2) hip joint
(3) knee joint

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

☐

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0180000043|!| ★

Which of the following statements about skeletal muscles is/are correct?

(1) Skeletal muscles work in pairs.
(2) When a skeletal muscle contracts, it becomes shorter and thicker.
(3) Skeletal muscles do not carry out involuntary actions.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1), (2) and (3)

☐

© 2020 Aristo Educational Press Ltd.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0180000044|!| ★

Which of the following is/are voluntary muscle(s)?

(1) eye muscles
(2) cardiac muscles
(3) biceps

A. (1) only
B. (3) only
C. (1) and (3) only
D. (1), (2) and (3)

□

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0180000045|!| ★

Which of the following muscles is/are ***not*** under conscious control?

(1) smooth muscles of the alimentary canal
(2) intercostal muscle
(3) cardiac muscle

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0180000046|!| ★

Which of the following are striated muscles?

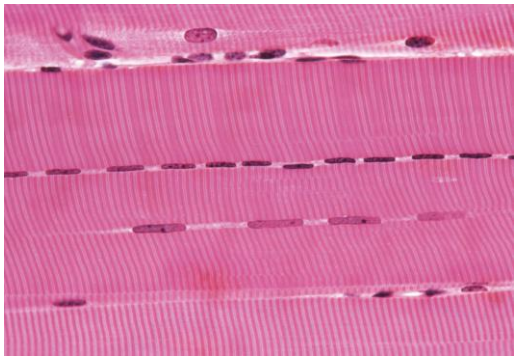
A. intercostal muscles
B. muscles in the stomach walls
C. ciliary muscles
D. erector muscles in the skin

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0180000047|!| ★★

The photomicrograph below shows some muscle fibres.



These muscle fibres are most likely found in

A. the heart.
B. the diaphragm.
C. the uterus.
D. the urinary bladder.

##B##

Skill assessed	Mark
Analysing data	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0180000048|!| ★

Which of the following are features of a muscle fibre?

(1) presence of multiple nuclei
(2) surrounded by membrane
(3) elongated in shape

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0180000049|!|

Smooth muscles can be found in the wall(s) of

(1) the heart.
(2) arteries.
(3) the urinary bladder.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMA0180000050|!| ★★

Figure 1 shows a man who is going to kick a football with his left light. Figure 2 shows bones and muscles in his left leg.

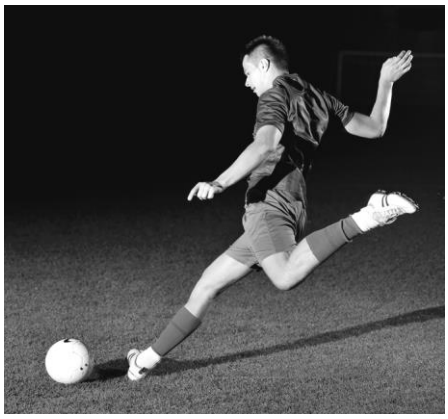


Figure 1

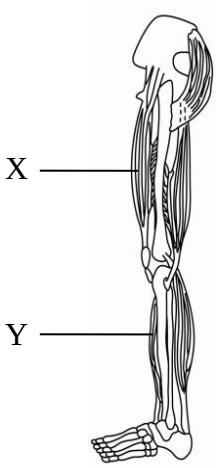


Figure 2

Which of the following correctly shows the states of muscles X and Y in his left leg?

	X	***Y***
A.	contracts	contracts
B.	contracts	relaxes
C.	relaxes	contracts
D.	relaxes	relaxes

##D##

Skill assessed	Mark
Applying concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMA0180000051|!| ★★

Figure 1 shows an athlete who has just started running. Figure 2 shows bones and muscles in his right leg.



Figure 1

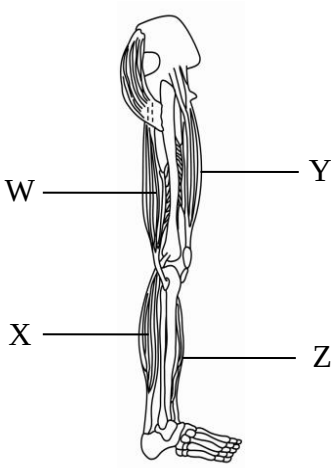


Figure 2

Which of the following correctly states the conditions of the leg muscles when the athlete straightens his right leg to produce the forward thrust?

	W	***X***	***Y***	***Z***
A.	contracts	contracts	relaxes	relaxes
B.	contracts	relaxes	relaxes	contracts
C.	relaxes	contracts	contracts	relaxes
D.	relaxes	relaxes	contracts	contracts

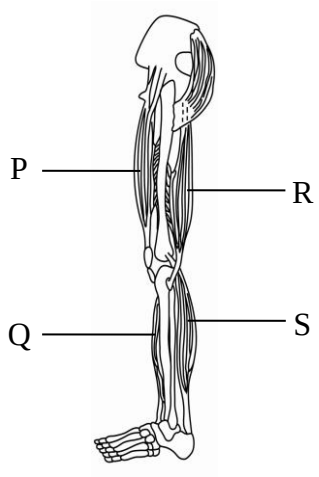
##C##

Skill assessed	Mark
Applying concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0180000052|!| ★

The diagram below shows the bones and muscles in a human leg.



Which of the muscles are extensors?

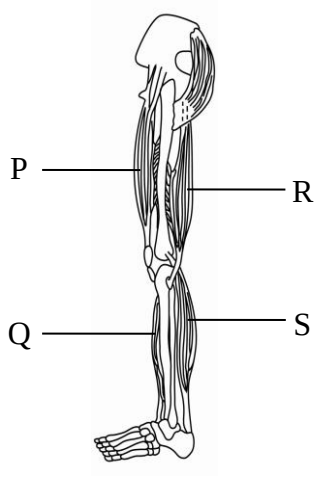
A. P and Q
B. P and S
C. R and Q
D. R and S

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0180000053|!|

The diagram below shows the bones and muscles in a human leg.



© 2020 Aristo Educational Press Ltd.

Which of the following is/are antagonistic muscle pair(s)?

(1) P and Q
(2) P and R
(3) Q and S

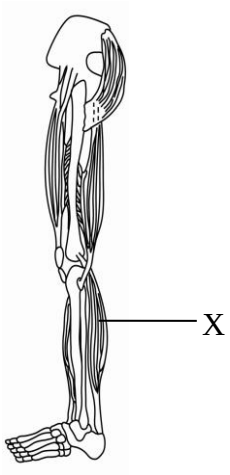
A. (1) only
B. (3) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0180000054|!| ★

The diagram below shows the bones and muscles in a human leg.



What will happen when muscle X contracts?

A. The leg will straighten at the knee.
B. The leg will bend at the knee.
C. The leg will straighten at the ankle.
D. The leg will bent at the ankle.

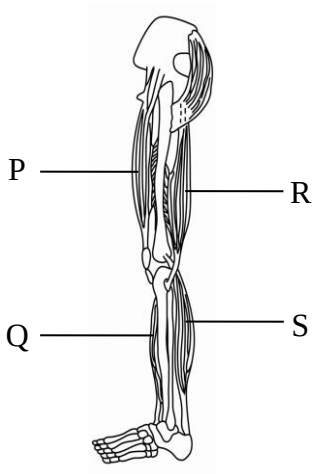
##C##

Skill assessed	Mark
Analysing data	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0180000055|!| ★

The diagram below shows the bones and muscles in a human leg.



Which of the muscles are flexors?

A. P and Q
B. P and S
C. R and Q
D. R and S

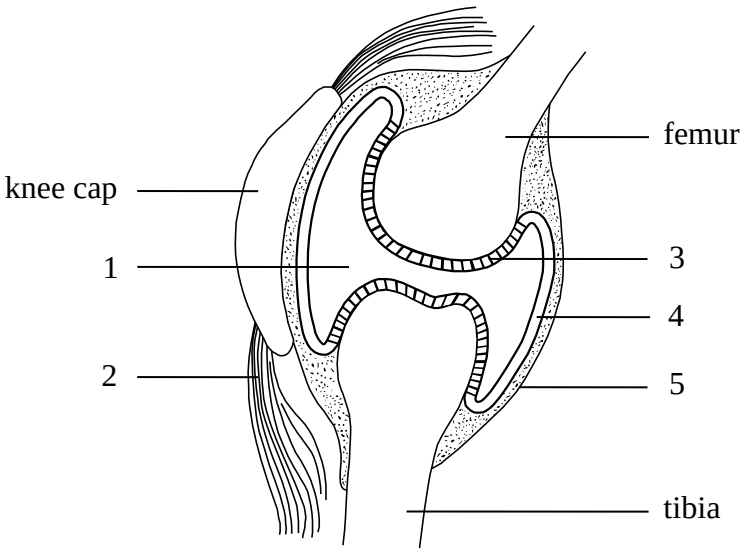
□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0180000056|!|

Directions: The following two questions refer to the diagram below, which shows a knee joint and its associated structures.



© 2020 Aristo Educational Press Ltd.

The fluid in cavity 1 is produced by

A. structure 3.
B. structure 4.
C. structure 5.
D. the bones in the joint.

□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0180000057|!|

What are structures 2 and 5?

	Structure 2	***Structure 5***
A.	ligament	tendon
B.	ligament	muscle
C.	tendon	ligament
D.	tendon	muscle

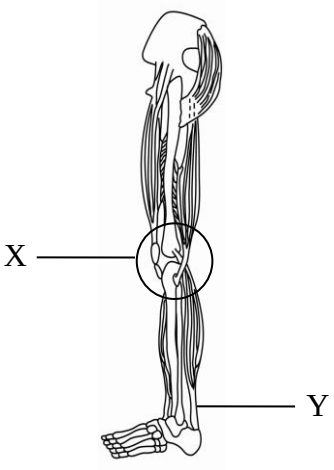
□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0180000058|!| ★

Directions: The following two questions refer to the diagram below, which shows the bones and muscles in a human leg.



© 2020 Aristo Educational Press Ltd.

Which of the following statements about joint X is ***incorrect***?

A. X is surrounded by ligaments to reduce the chance of dislocation.

B. The ends of the bones at X are covered by cartilage to reduce friction.

C. The ends of the bones at X are ball-shaped.

D. X allows the lower leg to swing forwards and backwards.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0180000059|!| ★

Which of the following correctly describes structure Y?

A. It is made up of dead cells.

B. It is hard and rigid.

C. It is inelastic.

D. It is able to contract.

##C##

Skill assessed	Mark
Analysing data	1

© 2020 Aristo Educational Press Ltd.